COMPACTNESS ACROSS HARDY EXPONENTS

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AI assistance was used in adapting and checking the argument and the exposition.

Brevig and Perfekt state: "$\mathscr{C}_\varphi : \mathscr{H}^2 \to \mathscr{H}^2$ is compact if and only if

$$\lim_{\operatorname{Re} w \to 1/2^+} \frac{\mathscr{M}_\varphi(w)}{\operatorname{Re} w - 1/2} = 0.$$

" The inferred question asks whether this characteristic-zero criterion is independent of finite $p \geq 2$. See Theorem 1.4.

Here $\mathscr{M}_\varphi(w) = \lim_{\sigma \downarrow 0} \lim_{T\to\infty} (\pi/T) \sum_{\varphi(s)=w,\, |\operatorname{Im} s|<T,\, \operatorname{Re} s>\sigma} \operatorname{Re} s$ is the mean counting function, with zeros counted with multiplicity, and φ has characteristic zero.

The Hilbert-space source is Brevig and Perfekt, *A mean counting function for Dirichlet series and compact composition operators*, Theorem 1.4, arXiv:2005.05094; the conditional local embedding is Chen, Fang, Guo, Hou, Shao, and Zhou, *The local embedding problem for Hardy spaces of Dirichlet series*, Theorem 1.1 and Proposition 10.1, arXiv:2609.11560v2.

Answered: yes, conditional for non-even p. Evidence: Sketch.

Conditional on: arXiv:2609.11560, Theorem 1.1 and Proposition 10.1.

The condition is necessary for every finite $p \geq 2$ and sufficient whenever the local embedding holds at p. Thus the equivalence is unconditional for even $p \geq 2$ and, conditional on the local-embedding theorem of Chen, Fang, Guo, Hou, Shao, and Zhou, for every finite $p \geq 2$.

Proof. Assume first that the local embedding holds at this p. Put $\nu = \varphi(+\infty)$ and $T_h f(s) = f(s+h)$. Horizontal translation is contractive on $\mathscr{H}^p$, and $T_h f \to f$ in norm as $h \downarrow 0$, by polynomial density. Since $p \geq 2$, the coefficient and evaluation estimates are

$$|a_n(f)| \leq \|f\|_p, \qquad |f(s)| \leq \zeta(2\operatorname{Re} s)^{1/2} \|f\|_p.$$

For $g = T_h f$, the Littlewood–Paley formula and Stanton's mean-counting identity give

$$\|C_\varphi g\|_p^p = |g(\nu)|^p + \frac{p^2}{2\pi} \int_{\mathbb{C}_{1/2}} |g(w)|^{p-2} |g'(w)|^2 M_\varphi(w)\, \mathrm{d}A(w). \tag{1}$$

Indeed, apply $\Delta |F|^p = p^2 |F|^{p-2} |F'|^2$ to $F = g \circ \varphi$ in the vertical mean formula, and change variables by the mean-counting function. The positive shift makes g, g' bounded on the half-plane of integration, so the truncations in that identity converge.

We need a weighted strip estimate. For $0 < \delta < 1$ and $1/2 < \theta \leq (1/2 + \operatorname{Re} \nu)/2$, the local embedding implies

$$\int_{1/2<\operatorname{Re} w<\theta} |f(w)|^{p-2} |f'(w)|^2 \frac{\operatorname{Re} w - 1/2}{|w-\nu|^{1+\delta}}\, \mathrm{d}A(w) \leq C \|f\|_p^p. \tag{2}$$

Here is the estimate. The local embedding on the boundary line transfers, by horizontal translation and polynomial approximation, to every line $x > 1/2$. Set $G(s) = f(s)(s + 1/2)^{-(1+\delta)/p}$, using the principal branch, and

$$m(x) = \int_{\mathbb{R}} |G(x+iy)|^p\, \mathrm{d}y.$$

Summing the unit-interval embedding bounds with the weight $|x + 1/2 + iy|^{-1-\delta}$ gives $m(x) \leq C\|f\|_p^p$ uniformly for $x > 1/2$. Differentiating under the integral and using the

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Laplacian identity shows that m is convex and

$$m''(x) = p^2 \int_{\mathbb{R}} |G(x+iy)|^{p-2} |G'(x+iy)|^2 \, \mathrm{d}y.$$

A bounded nonnegative convex function has $m'(x) \uparrow 0$; integrating this identity twice yields

$$\int_{\mathbb{C}_{1/2}} |G(w)|^{p-2} |G'(w)|^2 (\operatorname{Re} w - 1/2) \, \mathrm{d}A(w) \le C \|f\|_p^p.$$

Writing $f(w) = G(w)(w+1/2)^{(1+\delta)/p}$ and differentiating bounds the corresponding integral with denominator $|w+1/2|^{1+\delta}$. On the indicated strip, $|w+1/2| \le C_\nu |w-\nu|$, which proves (2).

The compactness condition and the Littlewood-type estimate in the proof of Brevig–Perfekt's Theorem 1.4 imply that for every $\varepsilon > 0$ there is $\theta > 1/2$ such that

$$M_\varphi(w) \le \varepsilon^2 \frac{\operatorname{Re} w - 1/2}{|w-\nu|^{1+\delta}}, \qquad 1/2 < \operatorname{Re} w < \theta. \tag{3}$$

Combining (1)–(3) controls the boundary strip by $C\varepsilon^2 \|f\|_p^p$. Away from the strip, the evaluation bound and Cauchy–Schwarz give $|g'(w)| \le C 2^{-\operatorname{Re} w}$. The remaining integral is finite because Stanton's formula at $p = 2$ applied to 2^{-s} gives $\int_{\mathbb{C}_{1/2}} |(2^{-w})'|^2 M_\varphi(w) \, \mathrm{d}A(w) < \infty$. This proves a bound for $C_\varphi T_h$ independent of h; letting $h \downarrow 0$ and using weak lower semicontinuity proves that C_φ is bounded on $\mathscr{H}^p$.

Now let (f_j) be a bounded weakly null sequence in $\mathscr{H}^p$. Its coefficients tend to zero, and the coefficient and derivative series show that $f_j, f_j' \to 0$ uniformly on every half-plane strictly to the right of $1/2$. More precisely, after choosing θ in (3), for all sufficiently large j,

$$|f_j(s)| \le \varepsilon, \qquad |f_j'(s)| \le \varepsilon |(2^{-s})'| \quad (\operatorname{Re} s \ge \theta).$$

The same bounds hold for $g = T_h f_j$, uniformly in $h > 0$, and $|g(\nu)| \le \varepsilon$. In (1), the part with $\operatorname{Re} w < \theta$ is at most $C\varepsilon^2$ by (2) –(3). On the rest, the integrand is bounded by $\varepsilon^p |(2^{-w})'|^2 M_\varphi(w)$, whose integral is finite by the preceding $p = 2$ estimate. Thus $\|C_\varphi T_h f_j\|_p^p \le C' \varepsilon^2$ uniformly for large j and all $h > 0$. Let $h \downarrow 0$ and use lower semicontinuity to get $\|C_\varphi f_j\|_p \to 0$. Reflexivity of $\mathscr{H}^p$ now implies compactness.

For necessity, write $\sigma = \operatorname{Re} w$ and use the normalized kernels

$$K_w(s) = \frac{\zeta(s+\overline{w})}{\zeta(2\sigma)^{1/2}}, \qquad G_{w,p}(s) = \frac{\zeta(s+\overline{w})^{2/p}}{\zeta(2\sigma)^{1/p}}.$$

Euler products give $\|K_w\|_2 = \|G_{w,p}\|_p = 1$ and $|K_w|^2 = |G_{w,p}|^p$. As $\sigma \downarrow 1/2$, every fixed coefficient of $G_{w,p}$ tends to zero; reflexivity therefore gives $G_{w,p} \rightharpoonup 0$. If C_φ is compact on $\mathscr{H}^p$, then $\|C_\varphi G_{w,p}\|_p \to 0$. The pointwise identity persists after composition, and the corresponding Hardy norms satisfy $\|C_\varphi K_w\|_2^2 = \|C_\varphi G_{w,p}\|_p^p$. Brevig–Perfekt's kernel estimate in the necessity proof of Theorem 1.4 gives

$$\frac{M_\varphi(w)}{\sigma - 1/2} \le C \|C_\varphi K_w\|_2^2,$$

so the stated mean-counting condition follows. The local embedding is unconditional for even p by the $p = 2$ case and integer powers; for other finite $p \ge 2$ it is conditional on the cited local-embedding preprint. □